

Independent yet Tight WCRT Analysis for Individual Priority Classes in Ethernet AVB

Jingyue Cao, Pieter J.L. Cuijpers, Reinder J. Bril, and Johan J. Lukkien
Eindhoven University of Technology
{J.Cao,P.J.L.Cuijpers,R.J.Bril,J.J.Lukkien}@tue.nl

ABSTRACT

In the high-tech and automotive industry, bandwidth considerations and widely accepted standardization are two important reasons why Ethernet is currently being considered as an alternative solution for real-time communication (compared to the traditional fieldbusses). Although Ethernet was originally not intended for this purpose, the development of the Ethernet AVB standard enables its use for transporting high-volume data (e.g. from cameras and entertainment applications) with latency guarantees.

The traditional busy-period analysis that ensures these latency guarantees relies on knowledge of all data streams, including interference. However, for complex industrial systems, the assumption that characteristics of all interfering data streams are known, is not realistic. As the amount of components connected to an Ethernet network grows, it becomes less likely that one can know or enforce the traffic patterns generated by all these components.

In this paper, we present an independent real-time analysis based on so-called eligible intervals, which does not rely on any assumptions on interfering priority classes other than those enforced in the Ethernet AVB standard. We prove our analysis is tight when no additional information on interference is known. Furthermore, we compare the results of our approach to the two most recent busy-period analyses, point out sources of pessimism in these earlier works, and discuss the advantages of using eligible-intervals over busy-period analysis when studying idling servers.

Keywords

Ethernet AVB; WCRT Analysis; Eligible Interval; Busy Period Analysis; Credit Based Shaping

1. INTRODUCTION

The increasing use of, for example, cameras in industrial systems, is resulting in an increasing demand for high-volume data transports. Especially in the high-tech and automotive industry, there is a need for high bandwidth

communication combined with the possibility to give real-time guarantees. Ethernet is currently being considered as a viable solution to this problem, as it provides a high bandwidth as part of a widely accepted standard. Originally, Ethernet was not intended for this purpose, but the development of the Ethernet AVB standard (from the Audio/Video Bridging task group [1]), enables the use of Ethernet for transporting high-volume data with latency guarantees. This standard relies on traffic shaping techniques, and in particular on the standardized credit-based shaper (see IEEE 802.1Qav [3]) to be able to guarantee required latencies and throughput for audio/video traffic, while preventing starvation of best-effort traffic.

One of the problems faced by the automotive industry, amongst others, is that there is a growing number of connected components present in modern cars and other vehicles. Those components are of a heterogeneous nature, in the sense that some pose hard real-time requirements on the communication network, while others only require soft real-time guarantees, or have no specified requirements at all. Furthermore, different components are often developed by different manufacturers, making it difficult to guarantee specific behavior in the traffic load the components put on the network.

To face the consequences of this heterogeneity, the transmission of frames in Ethernet AVB is scheduled in a prioritized fashion (see Figure 1). Streams of frames from several input ports are distributed over (FIFO) output queues depending on the priority class associated with the stream. In Ethernet AVB, several priority classes are possible, but in this paper we restrict ourselves to three: H (high priority), M (medium priority), and L (low priority / best-effort traffic), leaving the analysis of multiple classes as future work. Within class H and M , a stream of frames is shaped according to the credit-based shaping algorithm, constraining the transmission for those classes to a fraction of the available bandwidth (see Figure 2). This prevents starvation of the low priority levels, and in particular enables us to ensure latency guarantees for the M class, which is the main focus of this paper. For each class under shaping, the system keeps track of a credit value which decreases when a frame is transmitted and increases when there is a frame waiting or when credit is negative. In this way, even though there is a frame pending, a high priority class may find itself blocked at times, leaving some bandwidth for low priorities, thus improving the timing behavior of those low priorities.

Given the mechanisms of the Ethernet AVB standard, and the characteristics of frames and shapers, the worst-case re-

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

RTNS '16, October 19-21, 2016, Brest, France

© 2016 ACM. ISBN 978-1-4503-4787-7/16/10...\$15.00

DOI: <http://dx.doi.org/10.1145/2997465.2997493>

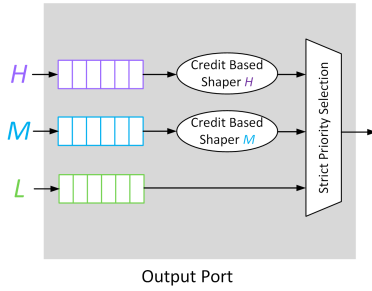


Figure 1: An output port of an Ethernet AVB switch. Priority classes H and M undergo credit-based shaping, while class L obtains the remaining bandwidth.

sponse time of a stream of frames can be bound using formal analysis techniques. Such analysis has been developed in e.g. [4, 5, 9, 11, 12, 14]. In [11], a timing analysis of Ethernet AVB is presented for a single scenario in which a single frame is interfered by both high- and low-priority traffic, but no formal study is executed to prove that the resulting formulas match the worst-case in all possible scenarios. In [14] an initial formula for worst-case response-times, based on Modular Performance Analysis (see [15]), is compared to simulations of a series of Ethernet AVB switches. In [9, 12] an extensive formal analysis of Ethernet AVB is given using the Compositional Performance Analysis approach [10], based on busy period analysis [6, 8, 13]. Finally, [4] and [5] independently improve on the work of [9]. In [5] the calculation of high-priority interference is carried out in more detail, while [4] addresses part of that same improvement, but then proceeds to consider multi-hop analysis.

The problem with these previous approaches, is that busy period analysis is developed for non-idling servers. Traditionally, it assumes a simultaneous arrival of interfering traffic, which in idling servers does not always lead to the worst-case scenario. This concern is not explicitly addressed in the works of [9], [4] and [5], but it is implicitly solved by simply adding the idling time as additional interference. Only in [5], it was realized that some interference is transmitted during this idling time. This interference should not contribute to the busy period. However, in this paper we show that the attempt in [5] to remove this source of pessimism is not entirely successful. By introducing eligible intervals (see [7]) we manage to simplify the analysis greatly, we explicitly take a non-simultaneous release into account, and we remove all sources of pessimism due to idling, while remaining independent of knowledge of the interference.

The goal of this paper is to perform a renewed analysis of Ethernet AVB, in which the analysis of the traffic in one priority class is independent of knowledge regarding the traffic in other classes. To do this, we extend our previous work on Ethernet AVB, found in [7]. In that paper, we introduced a new analysis technique that makes use of eligible intervals to determine a relation between interference, the built-up of credit in a credit-based shaper, and the relative delay of transmitted frames compared to a schedule without interference. We proved a delay by at most one interfering frame compared to the interference-free schedule in case of either low- or high-priority interference. However, the case of mixed interference has not been handled.

The main contribution of this paper is a formal analysis of

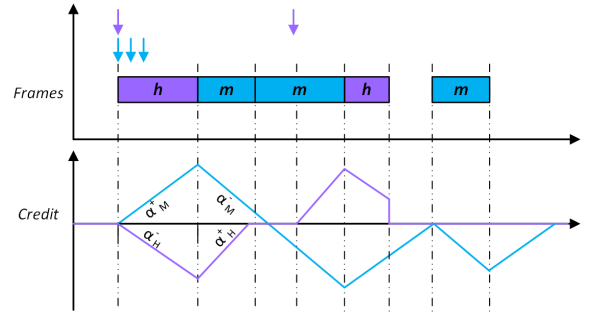


Figure 2: An example of credit-based shaping with class H and class M traffic. Arrows here indicate frame arrivals while rectangles indicate the transmission of frames. The transmission follows two rules: H is prioritized over M , and there must be non-negative credit at the start of a transmission. The rising slope of credit is set to the reserved bandwidth and the falling slope is set to the remaining bandwidth. For instance, if α_H^+ is set to 60 Mbps while the total available bandwidth is 100 Mbps, then α_H^- is set to 40 Mbps, which leads to a maximum utilization of H of 60%.

the medium-priority traffic in an Ethernet AVB switch with mixed interference. We study how the credit in a credit-based shaper builds up under the influence of a mix of unshaped low-priority interference, and shaped high-priority interference that is also subject to credit-based shaping. As it turns out, the study of eligible intervals still forms a good basis for this analysis, but the remaining proof needed a completely different approach. By studying the maximum amount of credit that can build up during interfering traffic, and taking into account the fact that high priority interference is also under shaping, we can produce a bound on the worst-case response time, and we are able to prove that this bound is tight, meaning that no better estimate can be given unless more is known regarding the interfering traffic. Finally, we compare our approach with the two most recent related analyses of Ethernet AVB, e.g. [4] and [5], and discuss the trade-off between knowledge of interference and pessimism of the analysis.

In the remainder of this paper, we start by describing our system model and our model of behavior of credit-based shaping in more detail. Subsequently, we recapitulate the most important parts of the analysis laid out in [7], and add our new analysis of the built-up of credit under mixed interference. Next, we discuss the tightness of our approach, and finally we compare our approach and results to those of [4] and [5].

2. SYSTEM MODEL

We are modeling the behavior of a single switch that is part of an Ethernet AVB network. The goal of this switch is to pass on frames from its inputs to its outputs according to some routing table. However, we are interested only in the time it takes to perform this task, and not in the actual routing of frames. In order to be able to calculate this, it is important to know how a switch schedules its transmissions.

As a basis, Ethernet AVB adopts non-preemptive strict priority scheduling. Frames are gathered in one queue for each priority class, and a frame can start its transmission only if there is no high priority queue that can start (strict

priority), if there are no earlier frames of the same priority in the queue (FIFO), and if there is currently no transmission going on (non-preemptive), see Figure 1.

On top of this, Ethernet AVB adds a credit-based shaping mechanism for the priority class to limit traffic bursts. In order to transmit a frame, the priority class may not have negative *credit*. Whenever credit is above or equal to 0, a frame may transmit (assuming the other rules also allow this), if credit is lower than 0, frames from low priority classes may transmit. Credit for a priority class X starts at 0, and while a frame is in transmission it drops at a rate of α_X^- . If credit is negative, no further frames of that class may be transmitted and credit rises with a rate of α_X^+ . Also, if class X has pending frames waiting but cannot transmit, credit rises at a rate of α_X^+ . Positive credit is reset to 0 if the queue does not contain any frames of class X . One example of credit-based shaping is illustrated in Figure 2.

In the standard, $0 < \alpha_X^+ \leq BW$ is set to correspond to the desired bandwidth reservation for class X , assuming a total bandwidth of BW . Subsequently, $\alpha_X^- = BW - \alpha_X^+$ is set to the remaining bandwidth. We observe that the maximum utilisation for class X is given by $\frac{\alpha_X^+}{BW}$, see also Figure 2. Finally, when designing a system it is reasonable to expect that the total reservation does not exceed the total available bandwidth, resulting in the requirement: $\alpha_H^+ + \alpha_M^+ \leq BW$. Although this requirement is strictly speaking not enforced by the standard, it seems a reasonable design choice in general, and we do need this assumption later in our analysis.

3. ANALYSIS

In our analysis, which largely follows the outline set up in [7], we focus on the behavior at the output port of an Ethernet AVB switch, as depicted in Figure 1. This means we assume that incoming frames have already been routed to the proper output port, and that the switch can freely transmit at its output. At the output port, frames are queued according to their class, resulting in three streams of frames, one for each class. We keep track of when frames of a certain stream arrive, and when they are transmitted, and we calculate how long a frame may remain in the output buffer. In other words, we are interested in the maximum amount of time between a frame entering the buffer and finishing transmission.

We first recap, in sections 3.1 to 3.3, the most important observations from [7], and analyze a few important cases in sequence. We start by assuming a single input stream of frames X for which we examine the effect of credit based shaping on the output stream by calculating the start times as the result of shaping. This results in an ordinary FIFO schedule, scaled to the reserved bandwidth. Next, we recall the analysis using so-called *eligible intervals* in which a stream has pending load available (i.e. there are frames to be sent) and also has non-negative credit (i.e. the frames can be sent unless the output port is otherwise occupied). Regardless of the interference, the start time of an eligible interval will always be smaller than or equal to the start time of its first frame in an un-interfered schedule. Then, based on the upper bound on the start time of each eligible interval, we further bound the relative delay of the interfered schedule compared to the un-interfered schedule. We show this delay is proportional to the credit at the start of transmission, and

as a consequence, we conclude that the maximum attainable credit bounds the maximum delay experienced by any frame compared to the un-interfered FIFO schedule.

The main contribution of this paper can be found in section 3.4, where we study the relation between the maximum attainable credit and the maximum frame sizes in the interfering traffic, and in this way we find a relatively simple expression for the worst-case time a frame may remain in the buffer. In section 4, we show that this estimate is in fact tight.

Definition 1 (Streams, frames, and timing). *A stream X is modeled as a finite set of frames $\{x_i \mid 1 \leq i \leq N_X\}$. For a frame x , we write $a(x) \geq 0$ and $s(x) \geq a(x)$ to denote its arrival time and transmission start-time, respectively. Furthermore, $C(x) > 0$ refers to the duration of the transmission (also called transmission time) of x . All frames in a stream X are assumed to be of the same priority class, which is also referred to as class X . We write $C_X^{max} = \max_{x \in X} C(x)$ to denote the maximum frame size in a stream X .*

Definition 2 (Credit of a class X). *The credit at time t associated with a class X is denoted by $CR_X(t)$. The credit evolves according to the rules of the shaper explained in the previous section. In particular, for $t \leq 0$ we have $CR_X(t) = 0$, because arrivals in a stream start at or after 0.*

Definition 3 (Pending load¹). *We call a frame x pending at time t when $a(x) \leq t$ and $s(x) + C(x) > t$. The pending load of x , $PL_x(t)$, is defined as $(C(x) - (t - s(x))) \min C(x)$ when x is pending; otherwise it is 0. The pending load of a stream X at time t , $PL_X(t)$, is the sum of the pending loads of its frames.*

Definition 4 (Eligible Interval; see also [7]). *An eligible interval $E = [E_s, E_e]$ for a stream X is an interval during which the stream has pending load and frames are eligible for transmission (i.e. there is non-negative credit, or a frame is actually transmitting). In other words, it is an interval for which the following criteria are met:*

- $CR_X(E_s) = 0$ and $PL_X(E_s) > 0$
- $CR_X(E_s - \epsilon) < 0$ or $PL_X(E_s - \epsilon) = 0$, for a vanishing small $\epsilon > 0$
- An X frame ends its transmission at E_e
- $CR_X(E_e) < 0$ or $PL_X(E_e) = 0$
- E_e is the smallest time strictly larger than E_s with these properties

The notion of eligible interval was introduced in [7], where we studied the behavior of Ethernet AVB for two types of interfering traffic: only low priority traffic and only (shaped) high priority traffic. In this paper, we address the more complex case in which interference may be a combination of the two. We re-use the result from [7] claiming that *the start-time of an eligible interval is always earlier than or equal to the start-time of its first frame in an un-interfered schedule*, and we also re-use the observation that *any experienced relative delay is bounded by the amount of credit at the*

¹Note, that in contrast to [6], we consider a frame that arrives at time t to contribute to the pending load at time t rather than immediately after time t .

start of transmission. However, we need a rather different approach to calculating this maximum credit in the mixed case. It is worth emphasizing, that in our analysis, streams are not restricted in the timing of the arrival of frames. A stream X or interfering stream Y may be periodic, sporadic, bursty, or have any other arrival pattern. In particular, it may represent the combination of several sources of traffic.

Eligible intervals have been labeled in grey context in all figures illustrating frame transmission. Observe, that transmission of frames in X only takes place during some eligible interval, meaning they are separated by intervals without transmission of X frames. Note also, that there is no idle time within an eligible interval, i.e. no time at which no transmissions take place at all. This makes the behavior within an eligible interval easier to analyse.

3.1 Shaping a single stream

We start with a single medium-priority stream M , and at this point assume there is no interference from other streams yet. In the following sections, we will also consider a high priority stream H and low priority stream L that model interference.

The first frame m_1 in this stream arrives at $a(m_1) \geq 0$ and the credit is initially 0. A frame can only start transmission when the credit is non-negative. Since positive credit is possible only when there is interference, frames start their transmission with a credit of 0. During the transmission, the credit decreases at a rate of α_M^- . The credit has to recover to zero before any next frame can be transmitted. The recovery time is the product of the frame duration and the ratio α_M^-/α_M^+ (see Figure 3). This gives us the following recurrence for start times.

Property 1. *Considering a single shaped stream M without interference, the start times are determined by*

$$s(m_1) = a(m_1)$$

$$s(m_{i+1}) = \left(s(m_i) + C(m_i) \cdot \left(1 + \frac{\alpha_M^-}{\alpha_M^+} \right) \right) \max a(m_{i+1})$$

Property 1 shows the shaping effect on a single stream to be the same as that of a basic FIFO schedule with enlarged execution times by a factor $(1 + \alpha_M^-/\alpha_M^+)$.

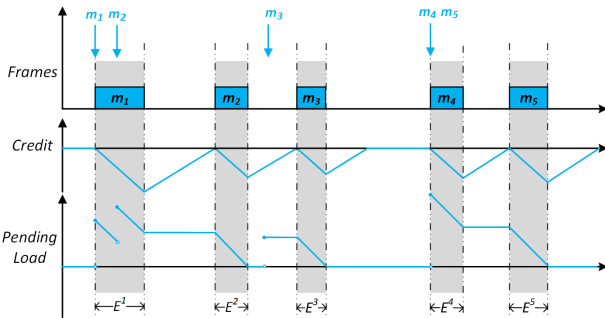


Figure 3: The transmission under credit-based shaping with only class M traffic. Arrows indicate the frame arrivals, and the transmission under credit based shaping is shown along with the evolution of the credit and pending load. Each eligible interval is labeled with E , within which there is only one M frame.

For a single un-interfered stream M , the intervals $E^i = [s(m_i), s(m_i) + C(m_i))$ represent a series of eligible intervals

of M . This is illustrated in Figure 3. Note, that when studying an un-interfered stream, we find that the shape of eligible intervals is rather simple. There is only one frame in each interval E , since the credit becomes negative when a frame is sent and no frame can transmit before the credit recovers to zero. Later, we see that in cases of interference, multiple frames may transmit in an eligible interval as well.

Our remaining analysis focusses on determining how interference may induce a deviation from the FIFO schedule described by Property 1.

3.2 The start of an eligible interval

In this section we study an arbitrary stream M , which may encounter interference from other low- or high-priority streams L and H . We will see that, due to interference, transmissions of frames in the stream M may cluster, and multiple frames can be transmitted in a single eligible interval. But even though the eligible intervals of the interfered stream are different compared to the un-interfered stream, we manage to relate the start-time of the eligible intervals in the interfered stream to the start-times of frames in the un-interfered stream.

For an interfered stream M , we write $s'(m)$ to denote the start time of m in the scenario with interference, while $s(m)$ is still used to denote the start time in the un-interfered scenario described in the previous section. We consider a single eligible interval E of M and recall the first of the two main theorems from [7], which states that the start E_s of the interval lies before or at the start time $s(m)$ (un-interfered scenario) of the first frame m transmitted in E . This allows us further on to relate the delay experienced by any frame back to the un-interfered schedule by just studying the interference within an eligible interval.

Theorem 1. *Let m be the first M frame transmitted in an eligible interval E , then $E_s \leq s(m)$ (i.e., E_s is at most the original start time of m).*

Proof. See [7]. □

3.3 Credit represents delay

Now that we have an upper bound on the start time of each eligible interval, we can study the relative delay of frames transmitted within an eligible interval. As it turns out, for those frames, the delay compared to the un-interfered scenario is proportional to the credit of the shaper at the start of transmission. Which gives us the second main theorem from [7].

Theorem 2. *Given a stream M subject to any interference, for each frame m we find $s'(m) \leq s(m) + \frac{CR_M(s'(m))}{\alpha_M^+}$.*

Proof. See [7]. □

3.4 Bounding the delay caused by interference

Following Theorem 2 we only need to prove a bound on the maximum credit that can build up in order to prove a bound on the delay in any possible scenario. The approach suggested in [7] turned out not to work for this, so from here on the analysis differs from our previous work. We find that the maximum delay is in fact determined by the maximum frame size occurring in the high and low priority streams.

Theorem 3. *Given a stream M interfered by a high-priority stream H and low-priority stream L , with M and H under credit based shaping. Assume $\alpha_M^+ + \alpha_H^+ \leq BW$. Then any frame in M can at most be delayed by $C_L^{max} \cdot (1 + \frac{\alpha_H^+}{\alpha_H^-}) + C_H^{max}$ compared to its un-interfered schedule.*

Proof. To prove this theorem, we only need to prove that CR_M is bounded by $(C_L^{max} \cdot (1 + \frac{\alpha_H^+}{\alpha_H^-}) + C_H^{max}) \cdot \alpha_M^+$, and subsequently use Theorem 2. To study the variation of the credit, we divide time into phases of three different types depending on what is in transmission: idling phase (nothing is in transmission), L phase (only L frames transmit) and MH phase (only M or H frames transmit). A phase only ends when another type of phase takes over.

Idling Phase

When nothing is in transmission, the credit of both streams, M and H , is negative (transmission is not allowed by shaper) or there is no pending load (positive credit is reset to 0). Hence, at any time t in an idling phase, the credit of M and H is non-positive: $CR_H(t) \leq 0 \wedge CR_M(t) \leq 0$.

L Phase

An L frame cannot transmit when either credit H or credit M is positive, so we find CR_H and CR_M are non-positive at the start of any L transmission. During the transmission, CR_H rises with a rate at most α_H^+ , and similarly CR_M rises with at most α_M^+ . Notice, that credit only rises above 0 if there are H or M frames pending, respectively. Once CR_H or CR_M rises above 0, no new L transmission can start, but the current one can finish. As a result, CR_H rises at most to $C_L^{max} \cdot \alpha_H^+$ and similarly CR_M rises to at most $C_L^{max} \cdot \alpha_M^+$. Hence, we conclude that at any time t in an L phase:

$$CR_H(t) \leq C_L^{max} \cdot \alpha_H^+ \wedge CR_M(t) \leq C_L^{max} \cdot \alpha_M^+.$$

MH Phase

The credit variation is more complex in an MH phase. Let t_s denote the start time of a given MH phase. To bound the credit of M at any time t ($t \geq t_s$) during this phase, we study the total (cumulative) time during which H transmits in $[t_s, t]$, and denote this by δ_H . Similarly, we write δ_M for the total time during which M transmits in this interval and we know that $\delta_H + \delta_M = t - t_s$. Our reasoning is mainly based on the credit variation rules and the reservation bound, i.e. $\alpha_M^+ + \alpha_H^+ \leq BW$. This bound implies $\alpha_M^+ \leq \alpha_H^-$ and $\alpha_H^+ \leq \alpha_M^-$, refer to Section 2 and further implies the essential inequality for the proof, $\alpha_M^+ \cdot \alpha_H^+ \leq \alpha_M^- \cdot \alpha_H^-$.

Clearly, in an MH phase, the credit of M can rise only when H is transmitting, and credit of H can rise only when M is transmitting. At time t , therefore, we know that:

$$CR_H(t) \leq CR_H(t_s) + \alpha_H^+ \cdot \delta_M - \alpha_H^- \cdot \delta_H$$

and

$$CR_M(t) \leq CR_M(t_s) + \alpha_M^+ \cdot \delta_H - \alpha_M^- \cdot \delta_M.$$

Furthermore, a new transmission cannot start when credit is negative, so credit cannot fall below the credit needed for transmitting one (maximum) frame:

$$CR_H(t) \geq -\alpha_H^- \cdot C_H^{max}.$$

Combining the above inequalities, we have:

$$CR_H(t_s) + \alpha_H^+ \cdot \delta_M - \alpha_H^- \cdot \delta_H \geq -\alpha_H^- \cdot C_H^{max}.$$

Now we compute the upper bound for $CR_M(t)$, and in the last step we use the inequality derived from the reservation bound:

$$\begin{aligned} CR_M(t) &\leq CR_M(t_s) + \alpha_M^+ \cdot \delta_H - \alpha_M^- \cdot \delta_M \\ &\leq CR_M(t_s) + \frac{\alpha_M^+}{\alpha_H^-} \cdot (CR_H(t_s) + \alpha_H^- \cdot C_H^{max} + \alpha_H^+ \cdot \delta_M) \\ &\quad - \alpha_M^- \cdot \delta_M \\ &= CR_M(t_s) + \frac{\alpha_M^+}{\alpha_H^-} \cdot CR_H(t_s) + \alpha_M^+ \cdot C_H^{max} \\ &\quad + (\frac{\alpha_M^+ \cdot \alpha_H^+}{\alpha_H^-} - \alpha_M^-) \cdot \delta_M \\ &\leq CR_M(t_s) + \frac{\alpha_M^+}{\alpha_H^-} \cdot CR_H(t_s) + \alpha_M^+ \cdot C_H^{max}. \end{aligned}$$

As we explained an MH phase only follows either an idling phase or an L phase. Therefore, the initial credits $CR_M(t_s)$ and $CR_H(t_s)$ are bounded by the results obtained in those cases, and we derive:

$$CR_M(t) \leq \left(C_L^{max} \cdot (1 + \frac{\alpha_H^+}{\alpha_H^-}) + C_H^{max} \right) \cdot \alpha_M^+,$$

which concludes our proof. $\square$

This concludes the proof of the main theorem of this paper. From this theorem it easily follows that given a worst-case response time of a class M frame in an un-interfered schedule $WR_{no-int}(m)$, the worst-case response time in an interfered schedule is bounded by:

$$WR_{int}(m) = WR_{no-int}(m) + C_L^{max} \cdot (1 + \frac{\alpha_H^+}{\alpha_H^-}) + C_H^{max}.$$

Using this insight, the WCRT analysis can be simplified to an analysis that considers only a single priority. In section 5, we'll give an example of an application of this.

As an added insight, we observe from the proof of Theorem 3 that during the combined execution of M and H frames, the total credit of both shapers decreases with at least a rate determined by their combined reservation.

Corollary 1. *Under the assumptions of Theorem 3, when either H or M frames are in transmission, the sum of the credit $CR_H(t) + CR_M(t)$ decreases at a rate of at least $BW - \alpha_M^+ - \alpha_H^+$. Consequently, at any time t we find $CR_H(t) + CR_M(t) \leq C_L^{max} \cdot (\alpha_H^+ + \alpha_M^+)$.*

Proof. Using the same definitions for t_s , t , δ_M and δ_H as in the proof of Theorem 3, we first observe that $\alpha_H^+ - \alpha_M^- = \alpha_M^+ - \alpha_H^- = -(BW - \alpha_M^+ - \alpha_H^+) \leq 0$, and continue to use observations from the previous proof to find:

$$\begin{aligned} CR_H(t) + CR_M(t) &\leq CR_H(t_s) + \alpha_H^+ \cdot \delta_M - \alpha_H^- \cdot \delta_H \\ &\quad + CR_M(t_s) + \alpha_M^+ \cdot \delta_H - \alpha_M^- \cdot \delta_M \\ &= CR_H(t_s) + CR_M(t_s) + (\alpha_H^+ - \alpha_M^-) \cdot \delta_M \\ &\quad + (\alpha_M^+ - \alpha_H^-) \cdot \delta_H \\ &= CR_H(t_s) + CR_M(t_s) - (BW - \alpha_M^+ - \alpha_H^+) \cdot (\delta_M + \delta_H) \\ &= CR_H(t_s) + CR_M(t_s) - (BW - \alpha_M^+ - \alpha_H^+) \cdot (t - t_s) \\ &\leq CR_H(t_s) + CR_M(t_s) \\ &\leq C_L^{max} \cdot (\alpha_H^+ + \alpha_M^+), \end{aligned}$$

which concludes the proof. $\square$

4. EXAMPLE AND PROOF OF TIGHTNESS

To illustrate the worst-case scenario that underlies Theorem 3, consider the eligible interval in Figure 4. This figure shows how a maximum credit for class M , and hence the maximum delay for a frame, can be achieved. The downward arrows indicate the arrival of frames, the solid rectangles indicate the execution of the combined H, M and L streams, the shaded rectangles indicate how the M -stream would execute without interference. The graphs below this indicate the total credit level of both shapers, and the individual credit levels of H and M during execution. In the worst-case scenario, the eligible interval starts with an L frame of maximum transmission time. Immediately after the start of L , the frame m_1 (our frame of interest) arrives together with a burst of H -frames. This burst fully depletes the H credit built up by the L transmission (indicated by a dot) and subsequently finishes with a maximum size frame so that the minimum credit of H is reached. After this, the frame m_1 is ready to transmit. At that point the credit of M has reached its maximum value, and frame m_1 has experienced its maximum delay compared to the un-interfered schedule. In the execution that follows, we see that the total credit keeps on decreasing, and that the maximum credit level of M is never obtained again. This is the essence of Corollary 1. Only in case $\alpha_H^+ + \alpha_M^+ = BW$, stream M may reach the maximum more than once, because the total credit remains the same.

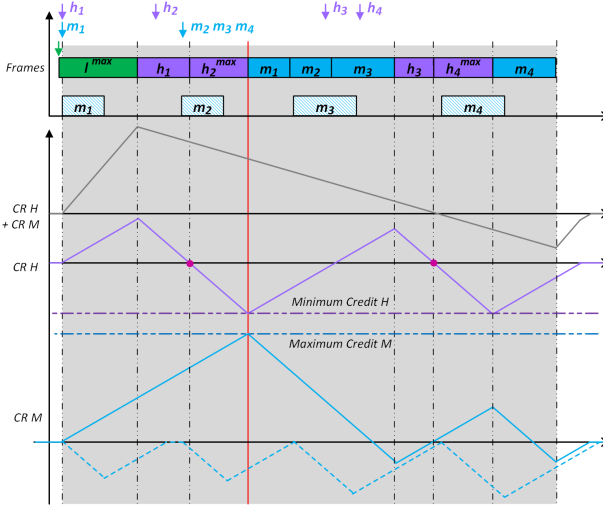


Figure 4: A stream of M interfered by a stream of H and L , and $\alpha_H^+ + \alpha_M^+ < BW$. The solid rectangles represent the transmission of streams H , M and L while the shaded ones represent the interference-free transmission of M . So do the solid and dashed lines representing the credit evolution. Frames of maximum transmission time have been labeled with superscript max , and the eligible interval has been labeled in grey context. The two dots on the curve of credit H emphasize the moment when credit reaches exactly zero and a frame of maximum size starts its transmission. The solid vertical line in red marks the timing when credit M reached the maximum value that we proved.

To prove that the formula developed in the previous sec-

tion gives a tight bound on the worst-case interference, it is enough to realize that the example illustrated in Figure 4 in fact provides a generic construction of the worst case. For any given medium-priority scenario, the arrival of a low-priority frame and immediately subsequent arrival of a number of high-priority frames can delay the transmission of any medium-priority frame by the amount specified in the formula. This construction can be carried out for any frame in a particular shaped stream, as long as we have the freedom to assign a worst case arrival time to a single low priority frame and have the freedom to inject a burst of high priority frames of the right size. Therefore, our approach is tight under the assumptions made in our system model.

5. COMPARISON WITH EARLIER WORK

In this section, we perform a comparison with the busy period analyses presented in 4 and 5. We aim at pointing out the sources of pessimism in those approaches that have been remedied by our use of eligible intervals. Furthermore, even though our own approach is tight in a setting where no detailed information regarding interfering traffic is assumed, it may be possible to remove some remaining pessimism when such information is added. In our comparison with 4 and 5 we also discuss the extent to which this additional information is helpful.

5.1 Illustrative example

The main theorem of this paper, is a relative result. It states that the worst-case response-time of any class M frame is tightly bound by its worst-case response-time in an un-interfered schedule plus a constant term determined by the maximum frame size of high and low priority traffic and the reservation of the high priority shaper. In order to compute the actual response-time of a frame, just knowing the relative delay compared to an un-interfered schedule is not sufficient. We also need to calculate the worst-case response-time of that un-interfered schedule. For this, it is necessary to make some assumptions on the class M traffic, which causes intra-class interference. Up to this point, such assumptions did not play a part in our system model. For the main result of this paper, it is not important how the un-interfered schedule is analysed, but when comparing our work to that of others, it is important that we have a way to do so.

As an arbitrarily chosen illustrative example, let us consider three sources of class M traffic: τ_1, τ_2, τ_3 . In line with the assumptions in 5, we assume that each source has periodic behavior characterized by the inter-arrival time T_i , and a maximum frame-size C_i . Note, that we only need to assume this for the class M traffic, not for the high- and low-priority sources of interference. For the latter, we only assume maximum frame sizes C_L^{max} and C_H^{max} , and the reservation of high priority α_H^+ . This is summarized in Table 1 where we assign values to those parameters. Finally, we assume a bandwidth $BW = 100$ Mbps in this table, and choose the respective reservations of the shapers to be 40% of this, so: $\alpha_H^+ = \alpha_M^+ = 40$ Mbps and $\alpha_H^- = \alpha_M^- = 60$ Mbps.

To calculate the response-time of the un-interfered M -priority schedule, we use the fact that this schedule behaves as a FIFO queue with added recovery times. Under the condition that the utilization of the sources is less than the reservation, i.e.: $\sum_{\tau_i \in M} \frac{C_i}{T_i} \leq \frac{\alpha_M^+}{BW}$, we can perform a simple

busy period analysis (see appendix) to find that the worst-case response-time for any frame from source τ_i is given by:

$$WR_{no-int}(\tau_i) = \left(\sum_{\tau_j \in M, j \neq i} C_j \cdot \left(1 + \frac{\alpha_M^-}{\alpha_M^+}\right) \right) + C_i.$$

Note, that the utilization condition is necessary and sufficient in order to guarantee that the un-interfered schedule has finite worst-case response times. Also note that the formula above is tight, because the worst-case scenario is simply achieved when all M sources arrive simultaneously. Then, adding the interference according to Theorem 3 gives us:

$$WR_{int}(\tau_i) = WR_{no-int}(\tau_i) + C_L^{max} \cdot \left(1 + \frac{\alpha_H^+}{\alpha_H^-}\right) + C_H^{max},$$

which for the example above returns the values in the right part of Table 1

Table 1: Example scenario, calculating worst-case response-times for class M sources τ_1, τ_2 and τ_3 using the analysis presented in this paper, given a total bandwidth of 100 Mbps. The parameters on priority levels are in grey.

	P	α^+ Mbps	C μs	C_i μs	T_i μs	WR μs
	H	40	1	-	-	-
τ_1	M	40	3	1	25	17.83
τ_2	M	40	3	3	30	14.83
τ_3	M	40	3	2	20	16.33
	L	-	2	-	-	-

5.2 Applying [4] and [5] to the same example

Now, in order to compare our approach with that of [4] and [5], we must first observe that in those approaches more knowledge regarding the interfering traffic is assumed than in ours. In particular, in [5] it is assumed that also the high priority traffic is characterized by a set of periodic sources. Moreover, a deadline associated with each high-priority task is used in the calculations of [5] that compensate for the fact that some high-priority traffic transmits during the recovery time. In [4], the Compositional Performance Analysis approach is used, meaning that high and medium priority traffic are characterized by arrival curves instead of as periodic sources. An arrival curve specifies the maximal and minimal amount of traffic in an arbitrary interval of a certain size. In this comparison, we follow the assumptions of [5], because it is a strictly stronger assumption than that of [4]. Any periodic source can easily be represented as a (periodic) arrival curve.

Potentially, the addition of information on the interference means that the worst-case performance bounds can be improved over ours, even though our approach is tight for our system model. Further on we will see examples when this is indeed the case, but also examples where both [4] and [5] are pessimistic.

It is outside the scope of this paper to repeat the entire algorithms presented by [4] and [5] here, but for completeness we must mention a small adjustment that is needed before we can compare the work of [5] to ours and that of [4]. We observe that the definition of worst-case response-time in [5] includes the recovery time after the transmission of interest, while in our approach and that of [4], the response-time

ends immediately after the transmission. In order to provide a fair comparison, we have (straightforwardly) adapted the equations (20) and (21) in [5] to reflect this. Also, in the second improvement of [5], the calculation of medium-priority recovery time introduces unnecessary pessimism, and we adapt it according to [4]. Furthermore, in [4], equation (18) contains a term I_{HPB} , used to bound the time during which credit of a shaper can build up before a burst. The exact way in which the authors calculate I_{HPB} , however, remains unclear from the paper. Therefore, we decided to use our own estimate of the maximum attainable credit instead.

Table 2 contains the results of applying the methods from [4] and [5] to our illustrating example. We refer to the online appendix of this paper for the Matlab code that was used to create this table, and the graphs further on (<http://www.win.tue.nl/~jcao/rtnscode/>). We have refined the information on high-priority tasks, by distinguishing two sources τ_4 and τ_5 , and we have added deadlines for both tasks. It can be noted immediately, that despite this added information, the results for this particular example are pessimistic compared to those in Table 1. This is surprising, because the worst-case scenario that goes with this example, displayed in Figure 5 shows that the burst of high-priority behavior is too small to fully deplete the credit that has been built up. The maximum credit for M is not reached and the worst-case response time predicted by our approach is thus not attained, because there is not enough flexibility to generate the necessary worst-case interference. Next, we set out to identify the sources of the pessimism in [4] and [5] that cause the over-estimation of the worst-case response time, and we suggest a solution to improve our own prediction in case details on the limited burst-size are known or computable.

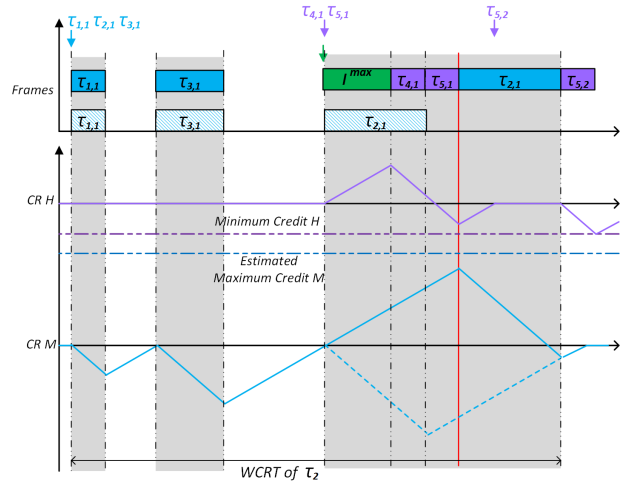


Figure 5: Worst-case scenario for τ_2 when adding knowledge regarding periodic interference (see Table 2). A maximum low-priority frame is released just before transmission of τ_2 is enabled, with subsequent arrivals of frames from high-priority τ_4 and τ_5 . Note how these two frames are insufficient to deplete the built-up credit of H , and prevent a built-up of the maximum credit of M , causing the transmission of τ_2 to start earlier than estimated.

5.3 Exploring pessimism

Table 2: Example scenario, calculating worst-case response-times for class M sources τ_1, τ_2 and τ_3 using the approaches in [4] and in [5], given a total bandwidth of 100 Mbps. The parameters on priority levels are in grey.

	P	α^+ Mbps	C μs	C_i μs	T_i μs	D_i μs	WR [4] μs	WR [5] μs
τ_4	H	40	1	1	10	10	-	-
τ_5	H	40	1	1	5	10	-	-
τ_1	M	40	3	1	25	-	23.5	18.5
τ_2	M	40	3	3	30	-	17.5	16.5
τ_3	M	40	3	2	20	-	20	19
-	L	-	2	-	-	-	-	-

The busy period analyses of [4] and [5] start out from the same basic analysis. Both papers initially identify a model in which four sources of interference are identified: high-priority interference, low-priority interference, medium-priority interference, and interference due to shaping. Incidentally this basic analysis in the two papers coincides for periodic sources, and for reference we have added the results of this analysis (under the name 'basic busy period analysis') in the figures that occur in this section.

Both [4] and [5] start by observing that the traditional calculation of high-priority interference, adding up all arrivals during a busy period, disregards the fact that the high priority is shaped. This source of pessimism is taken care of by adding a burst-rate model for the shaper.

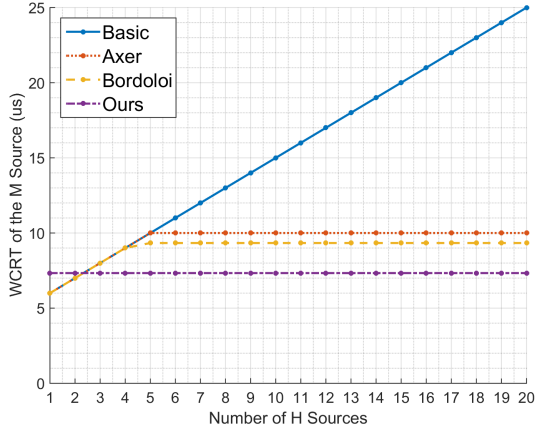


Figure 6: Worst-case response time of a medium-priority source given interference by n_H identical high-priority sources and a low-priority stream.

In Figure [6] we illustrate the influence of this improvement, by considering the same shaper reservations and maximum frame sizes as in the grey parts in Table [1]. We determine the worst-case response time of a single medium-priority source with $C = 3$, and $T = 30$, interfered by n_H identical high-priority sources with $C_i = 1$, $T_i = D_i = 5 \cdot n_H$, as well as a low-priority stream. Note, that we keep the total utilization of H at 20 Mbps (below its reservation α_H^+) as n_H ranges from 1 to 20.

The result of varying the number of high priority sources, is that the possible burst size of high priority traffic in-

creases. The figure clearly shows how, for low values of n_H , the burst size is smaller than the limits set by the credit based shaper. For those low values, the predictions of [4] and [5] turn out to be better than ours, because we do not consider the load generated by high priority sources. For higher values of n_H , the high priority shaping becomes dominant, and those of [4] and [5] eventually stabilize at a fixed value, while the basic busy period analysis keeps growing because it does not take high priority shaping into account at all. Our approach always remains the same value, which turns out to be tight and independent of the pattern of high priority traffic.

The reason why [5] performs better than [4] in Figure [6] is because of a second cause of pessimism, which was not addressed in [4]. This cause of pessimism was already briefly discussed in the introduction, when we mentioned that it is complicated (to say the least) to apply busy period analysis to idling servers. The four types of interference considered in the basic busy period analysis turn out not to be independent. If part of the high priority traffic transmits during a recovery period of the medium priority shaper, then this high priority traffic should actually not be counted as interference. This is because it does not contribute to the busy period. The authors of [5] realised this and have made an attempt to take this effect in to account.

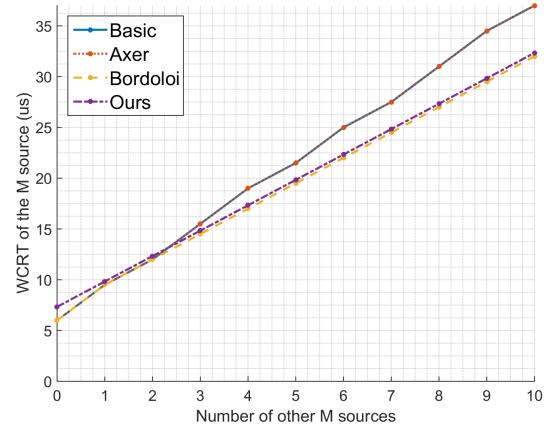


Figure 7: Worst-case response time of a medium-priority source given interference by 1 high-priority source, n_M medium priority sources, and a low-priority stream.

This is illustrated in Figure [7] in which we again consider the same shaper reservations and maximum frame sizes as in Table [1]. As before, we determine the worst-case response time of a single medium-priority source with $C = 3$, and $T = 30$, but this time the interference consists of a low-priority stream, a single high-priority source with $C = 1$, and $T = D = 5$, and n_M medium-priority sources with $C_i = 1$, $T_i = D_i = 5 \cdot n_M$.

By increasing the number of medium priority tasks, we increase the recovery time as well. Furthermore, by choosing a single high priority task with relatively small frame size, we ensure that the maximally built-up credit cannot be fully depleted. As a consequence, the approach of [5] is consistently better than ours in Figure [7] while the approach of [4] coincides with the basic busy period analysis and becomes more and more pessimistic as n_M grows.

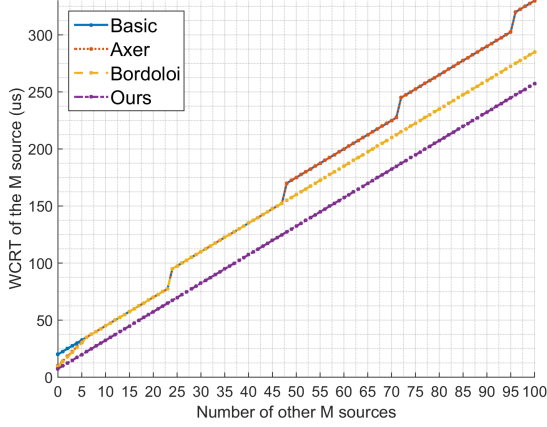


Figure 8: Worst-case response time of a medium-priority source given interference by 15 high-priority sources, n_M medium priority sources, and a low-priority stream.

Finally, if we adapt the scenario in which there are still n_M interfering medium priority sources, and consider $n_H = 15$ identical high priority sources instead of just 1, we see a completely different picture, shown in Figure 8. Now, our approach is tight again, because there is sufficient high priority traffic to deplete the credit to reach its minimum. Furthermore, if we range n_M from 1 to 100, we see that the approaches of [4] and [5] give better estimates than the basic analysis for low values ($n_M \leq 6$), since the shaping of high-priority is taken into account. For high values, the approach of [4] coincides with the basic analysis, and shows a ‘staircase’ behavior in its pessimism which indicates that the arrival of high priority interference is the limiting factor in the analysis. Interestingly, the analysis from [5] suffers from the same over approximation until $n_M = 47$, and only then merits from the implemented improvements.

The complexity of the approach set out in [5] makes it difficult to find a satisfactory explanation for the pessimism that still is present. Instead, it may be more promising to consider whether our own approach can be improved in cases where the burstiness of high priority traffic is insufficient to generate the worst-case behavior predicted by our approach.

5.4 Adding information

From the comparison so far, we conjecture that the only pessimism that remains in our approach occurs when the high priority traffic cannot produce a sufficiently large burst. Using an iterative approach similar to busy period analysis, we can calculate the maximum time *Burst* during which there can be a continuous transmission of high priority traffic. Considering the scenario in Figure 5, we claim the given knowledge of this burst our worst-case response-time analysis can be adapted to:

$$WR_{int}(\tau_i) = WR_{no-int}(\tau_i) + \min \left(C_L^{max} \cdot \left(1 + \frac{\alpha_H^+}{\alpha_H}\right) + C_H^{max}, C_L^{max} + Burst \right).$$

Obviously, the burstiness only makes a difference if *Burst* is smaller than time needed to deplete the built-up credit plus a single maximum high priority frame transmission.

Naturally, one can go even further, and consider the fact that the last transmission must be a frame of maximum size,

and that the transmissions preceding this must exactly fit the depletion time of the maximum credit. If those transmissions do not fit exactly (for instance, because frames have a fixed size) the worst-case can also not be attained. But such considerations only complicate the analysis, and are expected to lead to relatively small improvements compared to the results already achieved by our eligible interval analysis.

6. CONCLUSIONS AND FUTURE WORK

In this paper, we studied the worst-case response time of transmitting frames under credit-based shaping in an Ethernet AVB switch. We presented a technique to perform worst-case response time analysis, independent of knowledge of the interfering traffic (except those assumptions enforced by Ethernet AVB standard). We proved that our analysis yields a tight bound, meaning no better estimates can be obtained unless more is known of the interfering traffic.

Our comparison with the state-of-the-art (e.g. [4] and [5]) suggests that our approach is superior in situations where the burstiness of high priority traffic is large enough. In particular, the burst should be larger than the depletion time of the maximum credit built up by the low priority traffic plus a single maximum high priority frame transmission. Furthermore, given the input models assumed in [4] or [5], we conjecture it is fairly easy to estimate the maximum burst size and adapt our analysis slightly. This would give an equivalent result to the state at the art. However, such an improvement requires additional assumptions on the sources of interference, which we think is unreasonable considering the increasing number of connected components in a network. Furthermore, the improvements would complicate our algorithm, which is currently linear in complexity. Unlike busy period analysis, it does not rely on recursions of which the depth is data dependent.

We attribute the tightness as well as the simplicity of our approach to the use of eligible intervals, during which there is both pending load and credit available to a shaper. The use of these intervals leads to a separation of concerns in which we first relate eligible intervals to a known uninterfered schedule, and secondly relate the credit at the start of a transmission to the relative delay caused by the interfering traffic. Subsequently, we bound the relative delay by bounding the maximum attainable credit. We expect that this approach is extendible to having multiple (shaped) priority classes, and are currently investigating this further. Also we have hope that we will be able to use eligible intervals when studying other shaping strategies as well. For example, as future work we intend to study eligible intervals in the context of the Ethernet TSN standard [2], which is currently being developed. In Ethernet TSN other shapers are considered in combination with credit-based shaping to guarantee low latencies for control frames, adding another source of heterogeneity to the analysis. A third topic of investigation would be to see if the Compositional Performance Analysis approach that was used in [4] can be simplified using eligible intervals. In principle, if our eligible intervals can indeed be used to transform arrival curves into output curves, the study of multi-hop scenarios of idling servers becomes much easier.

Acknowledgements. This research is supported by the Dutch Technology Foundation STW under the Robust CPS program, project 12697.

7. REFERENCES

- [1] IEEE 802.1 Audio/Video Bridging Task Group.
- [2] IEEE 802.1 Time-Sensitive Networking Task Group.
- [3] IEEE Standard 802.1Qav-2009 - Forwarding and Queuing Enhancements for Time-Sensitive Streams.
- [4] AXER, P., THIELE, D., ERNST, R., AND DIEMER, J. Exploiting Shaper Context to Improve Performance Bounds of Ethernet AVB Networks. In *Proceedings of the 51st Annual Design Automation Conference (DAC)* (June 2014), pp. 1–6.
- [5] BORDOLOI, U., AMINIFAR, A., ELES, P., AND ZEBO, P. Schedulability analysis of Ethernet AVB switches. In *Embedded and Real-Time Computing Systems and Applications (RTCSA), IEEE 20th International Conference on* (Aug 2014), pp. 1–10.
- [6] BRIL, R., LUKKIEN, J., AND VERHAEGH, W. Worst-case response time analysis of real-time tasks under fixed-priority scheduling with deferred preemption. *Real-Time Systems* 42, 1 (2009), 63–119.
- [7] CAO, J., CUIJERS, P. J. L., BRIL, R. J., AND LUKKIEN, J. J. Tight worst-case response-time analysis for Ethernet AVB using Eligible Intervals. In *12th IEEE World Conference on Factory Communication Systems (WFCS)* (May 2016), pp. 1–8.
- [8] DAVIS, R., BURNS, A., BRIL, R., AND LUKKIEN, J. Controller Area Network (CAN) schedulability analysis: Refuted, revisited and revised. *Real-Time Systems* 35, 3 (2007), 239–272.
- [9] DIEMER, J., THIELE, D., AND ERNST, R. Formal worst-case timing analysis of Ethernet topologies with strict-priority and AVB switching. In *Industrial Embedded Systems (SIES), 7th IEEE International Symposium on* (June 2012), pp. 1–10.
- [10] HENIA, R., HAMANN, A., JERSAK, M., RACU, R., RICHTER, K., AND ERNST, R. System level performance analysis - the SymTA/S approach. *IEE Proceedings - Computers and Digital Techniques* 152 (March 2005), 148–166(18).
- [11] IMTIAZ, J., JASPERNEITE, J., AND HAN, L. A performance study of Ethernet Audio Video Bridging (AVB) for industrial real-time communication. In *Emerging Technologies Factory Automation (ETFA), 14th IEEE Conference on* (Sept 2009), pp. 1–8.
- [12] JONAS DIEMER, J. R., AND ERNST, R. Modeling of Ethernet AVB Networks for Worst-Case Timing Analysis. In *MATHMOD 2012 - 7th Vienna International Conference on Mathematical Modelling* (Vienna, Austria, Feb 2012).
- [13] LEHOCZKY, J. Fixed priority scheduling of periodic task sets with arbitrary deadlines. In *Proceedings of 11th Real-Time Systems Symposium* (Dec 1990), pp. 201–209.
- [14] REIMANN, F., GRAF, S., STREIT, F., GLAS, M., AND TEICH, J. Timing analysis of Ethernet AVB-based automotive E/E architectures. In *Emerging Technologies Factory Automation (ETFA), IEEE 18th Conference on* (Sept 2013), pp. 1–8.
- [15] WANDELER, E., THIELE, L., VERHOEF, M., AND LIEVERSE, P. System architecture evaluation using modular performance analysis: a case study. *International Journal on Software Tools for*

Technology Transfer 8, 6 (2006), 649–667.

APPENDIX

Lemma 1. *Given a single-priority stream X composed of periodic sources τ_i with maximum frame size C_i and period T_i , and given a credit-based shaper with reservation α_X^+ . If the total utilisation does not exceed the reserved bandwidth, i.e. $\sum_{\tau_i \in X} \frac{C_i}{T_i} \leq \frac{\alpha_X^+}{BW}$, then (in the absence of high- or low-priority interference) the worst-case response time for each single source is determined by:*

$$WR_{no-int}(\tau_i) = \left(\sum_{\tau_j \in X, j \neq i} C_j \cdot \left(1 + \frac{\alpha_X^-}{\alpha_X^+}\right) \right) + C_i.$$

Proof. The proof is based on a simple busy period analysis, in which we define $w(\tau_i, q)$ to be the longest possible waiting time, from the start of a busy period to the start of transmission of the q^{th} iteration of source τ_i . When a credit-based shaper has no high- or low-priority interference, credit can not build up. As we have seen in Property 1, the behavior is that of a FIFO queue with extended execution times. As a consequence, the worst-case behavior is obtained by considering a simultaneous release of all sources. We therefore find:

$$\begin{aligned} w(\tau_i, q) &= \sum_{\tau_j \in X, j \neq i} \left\lfloor \frac{(q-1)T_i}{T_j} + 1 \right\rfloor \cdot C_j \cdot \left(1 + \frac{\alpha_X^-}{\alpha_X^+}\right) \\ &\quad + (q-1) \cdot C_i \cdot \left(1 + \frac{\alpha_X^-}{\alpha_X^+}\right). \end{aligned}$$

Next, we can bound the response time of the q^{th} transmission by adding the maximum transmission time C_i of the q^{th} transmission and subtracting its arrival time. We then use the bound on the utilisation to derive:

$$\begin{aligned} r(\tau_i, q) &= w(\tau_i, q) - (q-1) \cdot T_i + C_i \\ &= \sum_{\tau_j \in X, j \neq i} \left\lfloor \frac{(q-1)T_i}{T_j} + 1 \right\rfloor \cdot C_j \cdot \left(1 + \frac{\alpha_X^-}{\alpha_X^+}\right) \\ &\quad + (q-1) \cdot C_i \cdot \left(1 + \frac{\alpha_X^-}{\alpha_X^+}\right) - (q-1) \cdot T_i + C_i \\ &\leq \sum_{\tau_j \in X, j \neq i} C_j \cdot \left(1 + \frac{\alpha_X^-}{\alpha_X^+}\right) + C_i \\ &\quad + (q-1) \cdot T_i \cdot \left(1 + \frac{\alpha_X^-}{\alpha_X^+}\right) \cdot \left(\sum_{\tau_j \in X} \frac{C_j}{T_j} - \frac{\alpha_X^+}{\alpha_X^+ + \alpha_X^-} \right) \\ &\leq \sum_{\tau_j \in X, j \neq i} C_j \cdot \left(1 + \frac{\alpha_X^-}{\alpha_X^+}\right) + C_i \\ &= r(\tau_i, 1). \end{aligned}$$

From this, we observe that the worst-case response time can be found in the first instance in the busy period, and obtain:

$$\begin{aligned} WR_{no-int}(\tau_i) &= \max_{q \geq 1} r(\tau_i, q) \\ &= r(\tau_i, 1) \\ &= \sum_{\tau_j \in X, j \neq i} C_j \cdot \left(1 + \frac{\alpha_X^-}{\alpha_X^+}\right) + C_i. \end{aligned}$$

□